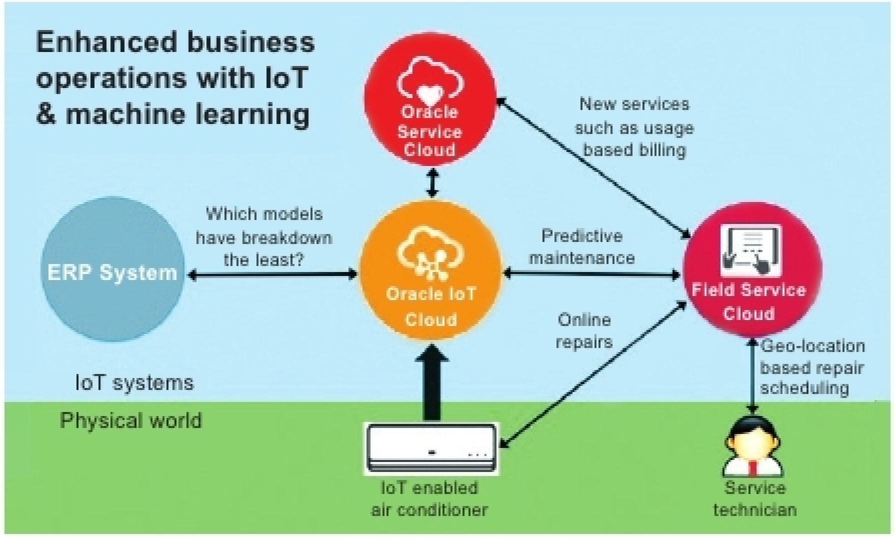

Figure 2: Increase in business operations with machine learning in IoT (Image source: *googleimages.com*)

unstructured data. But when it comes to IoT, it is often necessary to identify the correlations between dozens of sensor inputs and various external factors which are producing millions of data points. We know that traditional data analysis needs a model that is built on past data and also an expert opinion to establish a relationship between different variables. When it comes to ML, it directly starts with the outcome variables and then automatically looks for different predictor variables and their interactions. Hence, ML is really valuable when we know what we want but do not know the important input variables to come to that decision. Different ML algorithms learn from the sets of data that are important in achieving that goal.

2. Machine learning is quite valuable for accurately predicting future events. All the different data models that are built using traditional data analytics are static, but various ML algorithms constantly improve over time as more data is captured and assimilated. This indicates that the used algorithms can make predictions, compare against their past predictions, see what actually happens and then adjust to become more accurate.
3. The predictive capabilities of ML are extremely useful in industrial settings. By drawing the data from multiple sensors present in or on machines, different ML algorithms can actually learn whatever is typical for the machine and also detect whenever something abnormal starts occurring.
4. Machine learning helps to predict when a device connected to the IoT needs maintenance; this is incredibly valuable, translating into millions of dollars in saved costs. For example, Goldcorp, a mining company, is now using ML to make predictions with over 90 per cent accuracy about when maintenance is required, hence cutting costs considerably.
5. We all know that Amazon and Netflix use ML techniques to figure out our preferences in order to provide better experiences for us, suggesting different products that we might like and even provide relevant recommendations for movies and TV shows. In the case of IoT, ML can be extremely valuable in shaping our environment to suit our personal preferences. For example, Nest Thermostat uses ML to learn how hot or cool we like our homes to be, ensuring that the house is at the right temperature when we get home from work.
6. In any IoT situation, ML can really help different companies take the billions of data points that they have and distil these down to whatever is really meaningful to them.
7. To analyse any set of data from the IoT in real-time (in order to accurately identify the previously known and the never-before seen new patterns), all machines capable of generating and then aggregating such Big Data are allowed to learn normal behaviour using ML and uncover anything outside the norm.

Major ML techniques implemented in IoT

***Artificial neural networks*:** This type of learning technique is also called neural networking. It is basically a learning algorithm inspired by the structure and functional aspects of biological neural networks. Various computations are actually arranged in terms of an interconnected group consisting of artificially designed neurons. This helps to process the information using the connectionist approach. All different modern neural networks are non-linear statistical tools, which are used for data modelling. They are basically used to model the complex relationships between inputs and outputs, to find the patterns in the data.

K-means algorithms: This addresses a widely used node clustering problem as it has linear complexity and also a simple implementation. The K-means actually steps in to resolve different node clustering problems. It randomly chooses K nodes to be initial centroids for the different clusters. It labels each of the nodes with its closest centroid using a distance function. Then it re-computes the centroids using current node memberships. It stops if the convergence condition is found to be valid; else, it cycles back to the previous step.

***Inductive logic programming (ILP)*:** This is a method used to rule the learning system with the help of logical programming as its representation for different input examples, background knowledge and the hypotheses. If we have the encoding of any known background knowledge with specific sets of examples that represent a logical database of the facts, then a hypothesised logic program can be derived using an ILP system. This entails all the positive and no negative instances, and accepts any programming language for representing the hypotheses.

Clustering: This is a technique used for the assignment of different observations into various clusters (sets) so that all the observations present within the same cluster